

Supplementary Information for “Riemann zeros from Floquet engineering a trapped-ion qubit”

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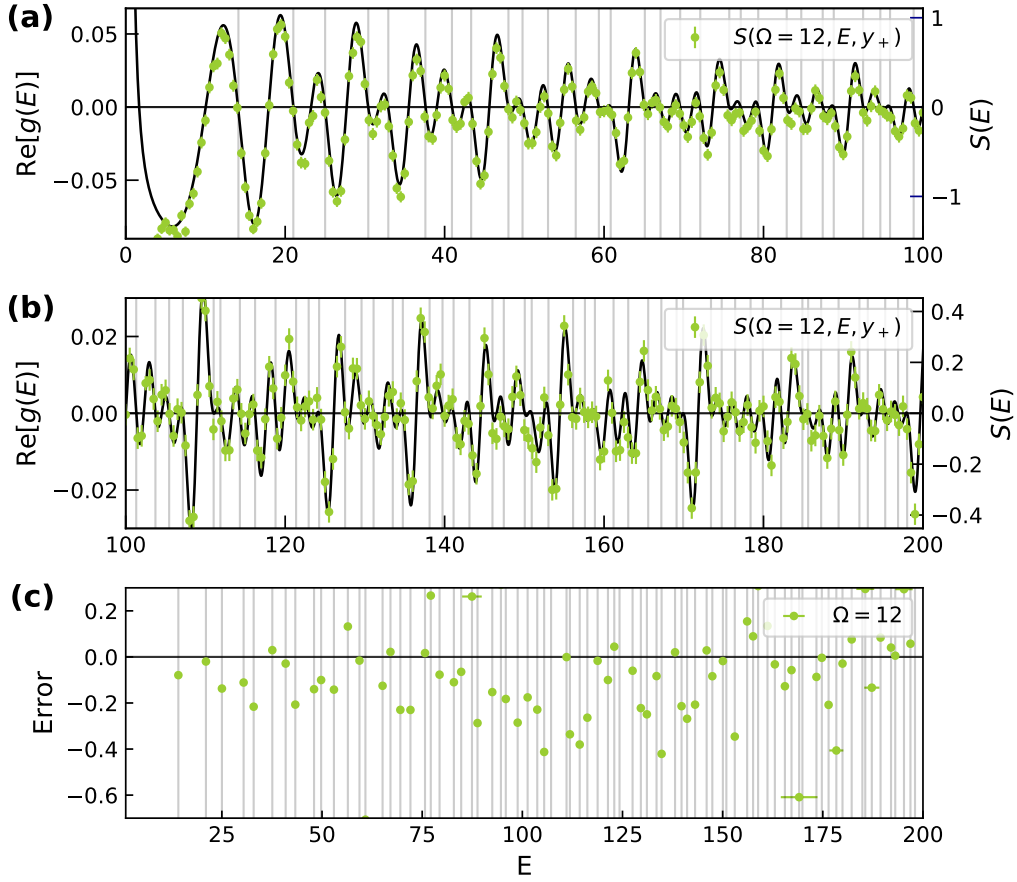
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Supplementary Figure 1. Identifying Riemann zeros by observing the frozen dynamics of the state for $\Omega = 12$. In (a) and (b), the black curve is the real part of $g(E)$. The vertical grey lines indicate the Riemann zeros, e.g., $\{14.1347, 21.022, 25.0109, 30.4249, 32.9351, \dots\}$. The other points where $g(E)$ crosses the zero axis correspond to the real part of the zeta function vanishing but not the imaginary part, and so do not represent Riemann zeros. The dots are the S parameter which is defined as $S(\Omega, E, y_+) = \sum_n [P(n, \Omega, E) - 1/2]$, where $n = \{5, 10, 15, 20, 25, 30\}$. $S(\Omega, E, y_+)$ is used to identify the Riemann zeros by observing $S = 0$. $S(\Omega = 12, E, y_+)$ shows excellent agreement with $g(E)$ for both (a) $0 \leq E \leq 100$ and (b) $100 \leq E \leq 200$ and thus acts as a good indicator of the location of the Riemann zeros. The first 80 Riemann zeros can be identified with high accuracy. The measurement can be extended to higher E without loss of efficiency and accuracy. Data points with $E \leq 100$ ($E \geq 100$) were obtained by 2000 (5000) measurements. The error bar δS of $S(\Omega, E, y_+)$ is the sum of the statistical errors of the corresponding $P(n, \Omega, E)$, where $n = \{5, 10, 15, 20, 25, 30\}$, within one standard deviation. (c) Zeros were extracted by interpolating $S(\Omega, E, y_+)$ using a cubic polynomial 4000 times. Each time, S is sampled randomly in $[S - \delta S, S + \delta S]$. The extracted zeros are the mean values of the interpolated zeros (see Supplementary Table 1 and Supplementary Table 2). The error (dot) is the difference between the extracted zeros and the exact zeros, with the error bars indicating the standard deviations of the mean values in the interpolation.

No.	Exact	$\Omega = 5$	$\Omega = 8$	$\Omega = 12$	$\Omega = 16$
1	14.135	14.07(1)	14.06(2)	13.99(4)	14.03(3)
2	21.022	21.04(2)	21.00(2)	20.93(5)	20.82(3)
3	25.011	24.70(3)	24.87(2)	24.87(7)	24.99(4)
4	30.425	30.59(2)	30.31(2)	30.29(3)	30.27(4)
5	32.935	32.76(3)	32.72(3)	32.57(8)	32.29(23)
6	37.586	37.64(2)	37.62(2)	37.39(2)	37.59(4)
7	40.919	40.95(2)	40.89(3)	40.78(3)	40.70(4)
8	43.327	42.85(9)	43.12(4)	43.23(4)	42.74(40)
9	48.005	48.23(4)	47.87(6)	47.94(6)	47.75(9)
10	49.774	49.26(19)	49.67(3)	49.36(23)	49.23(22)
11	52.970	52.93(2)	52.83(4)	52.88(5)	52.78(5)
12	56.446	56.56(3)	56.58(3)	56.28(3)	56.49(5)
13	59.347	59.44(5)	59.33(9)	59.35(6)	59.08(28)
14	60.832	60.10(48)	60.13(414)	60.41(144)	60.67(9)
15	65.113	65.53(11)	64.99(4)	65.05(6)	64.92(6)
16	67.080	67.06(5)	67.10(3)	66.98(10)	66.50(40)
17	69.546	69.36(4)	69.32(7)	69.11(28)	69.44(7)
18	72.067	71.82(3)	71.84(3)	71.76(8)	71.95(7)
19	75.705	76.33(37)	75.72(12)	75.23(332)	75.35(317)
20	77.145	76.84(8)	77.41(6)	76.80(9)	76.82(83)
21	79.337	78.89(6)	79.26(2)	78.95(4)	79.09(9)
22	82.914	84.12(117)	82.80(2)	82.67(4)	82.74(8)
23	84.736	84.82(2)	84.67(3)	84.31(12)	84.58(8)
24	87.425	87.50(7)	87.69(244)	87.23(6)	87.20(8)
25	88.809	87.93(47)	88.52(3)	88.33(47)	88.70(128)
26	92.492	92.96(4)	92.34(3)	92.37(5)	92.24(6)
27	94.651	94.55(59)	94.97(12)	94.34(144)	94.34(211)
28	95.871	95.82(8)	95.69(5)	94.66(290)	95.13(125)
29	98.831	98.87(3)	98.55(6)	98.69(4)	98.74(6)

Supplementary Table I. Comparison of the experimentally measured Riemann zeros with the true values ($1 \leq E \leq 100$) for different driving frequencies. Zeros were extracted by interpolating $S(\Omega, E, y_+) \pm \delta S$ with a cubic polynomial 4000 times, where δS is the $1 - \sigma$ standard deviation of S . Each time, S is sampled randomly in $[S - \delta S, S + \delta S]$. The zeros are the means of the interpolated zeros. The values in parentheses denote the standard deviation of the means in terms of the least significant digit.

No.	Exact	$\Omega = 8$	$\Omega = 12$	$\Omega = 16$
30	101.318	101.13(3)	101.31(3)	101.33(5)
31	103.726	103.50(4)	103.74(9)	103.66(5)
32	105.447	105.03(16)	105.49(7)	104.46(58)
33	107.169	106.03(112)	106.99(17)	106.93(5)
34	111.030	111.03(16)	111.60(227)	110.27(173)
35	111.875	111.54(19)	111.88(7)	112.86(337)
36	114.320	113.94(10)	114.49(9)	114.06(6)
37	116.227	115.96(11)	116.12(6)	115.82(20)
38	118.791	118.77(3)	118.71(4)	118.96(7)
39	121.370	121.27(6)	121.78(97)	122.26(69)
40	122.947	122.99(22)	123.48(323)	123.12(5)
41	124.257	122.82(48)	124.43(19)	123.85(38)
42	127.517	127.46(3)	127.51(4)	127.36(7)
43	129.579	129.36(32)	129.66(12)	129.57(12)
44	131.088	130.84(16)	130.91(156)	131.22(5)
45	133.498	133.41(7)	134.00(36)	133.62(12)
46	134.757	134.34(24)	134.89(132)	134.45(27)
47	138.116	138.14(8)	138.27(10)	138.12(11)
48	139.736	139.52(6)	139.98(9)	139.72(9)
49	141.124	140.86(8)	141.20(5)	140.57(107)
50	143.112	142.90(9)	142.72(22)	142.91(13)
51	146.001	146.03(8)	145.80(4)	146.24(52)
52	147.423	147.34(8)	147.17(46)	147.43(11)
53	150.054	150.04(6)	149.62(73)	150.10(125)
54	150.925	150.20(414)	149.47(197)	150.96(22)
55	153.025	152.68(6)	152.79(5)	152.89(14)
56	156.113	156.27(10)	156.02(203)	156.19(343)
57	157.598	157.69(20)	157.60(89)	157.40(15)
58	158.850	159.16(379)	158.84(89)	158.57(79)
59	161.189	161.32(8)	161.00(4)	161.25(3)
60	163.031	163.00(5)	162.43(58)	162.75(12)
61	165.537	165.41(5)	165.94(26)	165.71(14)
62	167.184	167.13(5)	167.08(11)	167.42(85)
63	169.095	168.49(458)	169.16(49)	169.01(14)
64	169.912	169.18(67)	169.17(27)	169.80(138)
65	173.412	173.33(5)	173.60(19)	173.36(8)
66	174.754	174.75(3)	174.65(7)	174.40(105)
67	176.441	176.23(11)	176.52(16)	176.41(13)
68	178.377	177.97(175)	178.26(10)	178.11(11)
69	179.916	179.89(5)	180.01(262)	179.36(51)
70	182.207	182.28(6)	182.14(6)	181.98(8)
71	184.874	185.51(294)	184.82(17)	184.77(86)
72	185.599	185.89(223)	185.43(24)	184.60(328)
73	187.229	187.10(188)	187.04(72)	187.22(34)
74	189.416	189.50(0)	189.28(6)	189.23(6)
75	192.027	192.07(9)	192.20(169)	192.42(41)
76	193.080	193.09(4)	193.10(12)	193.06(159)
77	195.265	195.56(217)	195.18(80)	195.55(58)
78	196.876	196.93(28)	196.04(304)	196.81(9)
79	198.015	196.81(42)	197.74(10)	197.80(154)
80	201.265	199.24(576)	200.14(5)	200.47(16)

Supplementary Table II. Comparison of the experimentally measured Riemann zeros with the true values ($100 \leq E \leq 200$) for different driving frequencies. Zeros were extracted by interpolating $S(\Omega, E, y_+) \pm \delta S$ with a cubic polynomial 4000 times, where δS is the $1 - \sigma$ standard deviation of S . Each time, S is sampled randomly in $[S - \delta S, S + \delta S]$. The zeros are the means of the interpolated zeros. The values in parentheses denote the standard deviation of the means in terms of the least significant digit.